



UNIVERSITÄT PADERBORN
Die Universität der Informationsgesellschaft

**Faculty of Electrical Engineering,
Information Technology and Mathematics (EIM)**

Explainable AI Mini Project

Project Group Report

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1 Abstract

Resource Description Framework (RDF) knowledge graphs represent information as interconnected entities and relationships, making them valuable for many machine learning applications. However, directly interpreting predictions generated from graph-structured data can be challenging. This project investigates the application of Explainable Artificial Intelligence (XAI) techniques to explain the predictions of a machine learning model trained on the AIFB RDF dataset, where the task is to classify researchers into their corresponding research groups. The RDF knowledge graph was first analyzed and converted into a structured tabular dataset through a feature engineering process that preserved semantically meaningful information while avoiding target leakage. Several classical machine learning models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, were trained and evaluated using the original training and test split. The Random Forest classifier was selected for explanation due to its competitive predictive performance. To improve the transparency of the model, multiple explainability techniques were applied, including built-in feature importance, permutation feature importance, Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE) plots, and SHAP. The generated explanations consistently identified semantic research topics and publication-related features as the most influential factors affecting the model’s predictions. Furthermore, local SHAP explanations provided interpretable insights into both correctly classified and misclassified researchers. Overall, the project demonstrates that combining RDF feature engineering with explainable machine learning enables transparent and meaningful interpretation of classification decisions on knowledge graph data.

2 Introduction

2.1 Background

Knowledge graphs are widely used to represent structured information by connecting entities through different relationships. One common way of representing a knowledge graph is the Resource Description Framework (RDF), where information is stored as subject–predicate–object triples. RDF datasets are used in many application areas such as semantic web technologies, recommendation systems, digital libraries, and scientific knowledge bases.

In recent years, machine learning models have been applied to RDF knowledge graphs for tasks such as node classification, link prediction, and entity classification. Although these models can achieve good predictive performance, they often provide little information about why a particular prediction was made. Understanding the reasoning behind a prediction is especially important when the model is used to support human decision making.

Explainable Artificial Intelligence (XAI) addresses this problem by providing methods that help users understand the behaviour of machine learning models. Different explanation techniques can identify the most important features, show how predictions change when feature values change, and explain the reasons behind individual predictions. These explanations increase the transparency of the model and help users evaluate whether the model is making reasonable decisions.

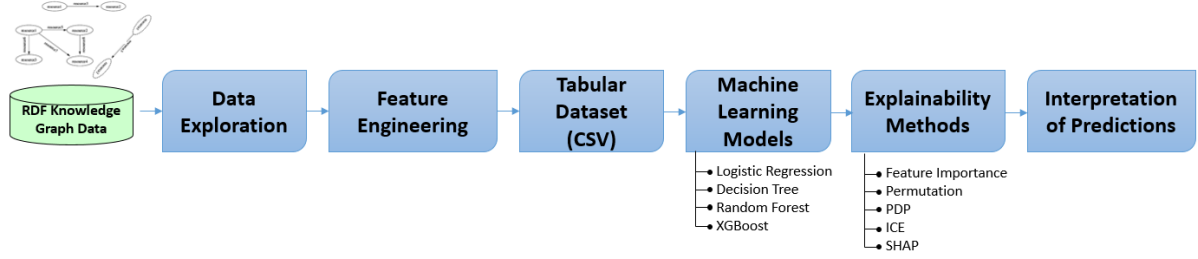


Figure 1: Overall workflow of the proposed Explainable AI pipeline.

2.2 Objective

The objective of this project is to explain the predictions of a machine learning model trained on an RDF knowledge graph. The AIFB dataset was selected for this work because it provides information about researchers, publications, research topics, projects, and research groups. The prediction task is to classify each researcher into the correct research group.

Since most classical machine learning algorithms require tabular data, the RDF graph was first analysed and converted into a structured feature matrix through a feature engineering process. During this process, only meaningful features were selected, and features that could introduce target leakage were carefully avoided.

After constructing the feature matrix, multiple machine learning models were trained and evaluated using the original train-test split provided with the dataset. Finally, several explainability techniques from the course, including Feature Importance, Permutation Feature Importance, Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE) plots, and SHAP, were applied to understand the behaviour of the selected model.

2.3 Overall Workflow

Figure 1 summarises the complete workflow followed in this project. The work started with analysing the RDF knowledge graph and understanding the available entities and relationships. Based on this analysis, a set of meaningful features was designed and converted into a tabular dataset. Different machine learning models were then trained and compared using the original training and testing split. Finally, explainability techniques were applied to analyse both the overall behaviour of the model and the reasons behind individual predictions.

3 Data Analysis and Feature Engineering

3.1 Dataset Overview

Table 1: Summary statistics of the AIFB RDF dataset.

Statistic	Value
RDF Triples	29,226
Entities	3,187
Predicates	47
URL Objects	20,858
Literal Objects	8,368

The AIFB dataset is an RDF knowledge graph describing researchers, publications, research topics, projects and research groups. The objective of this project is to predict the research group of each researcher. Since RDF data is represented as triples rather than a tabular structure, feature engineering is required before classical machine learning algorithms can be applied.

3.2 RDF Graph Analysis and Person Neighbourhood

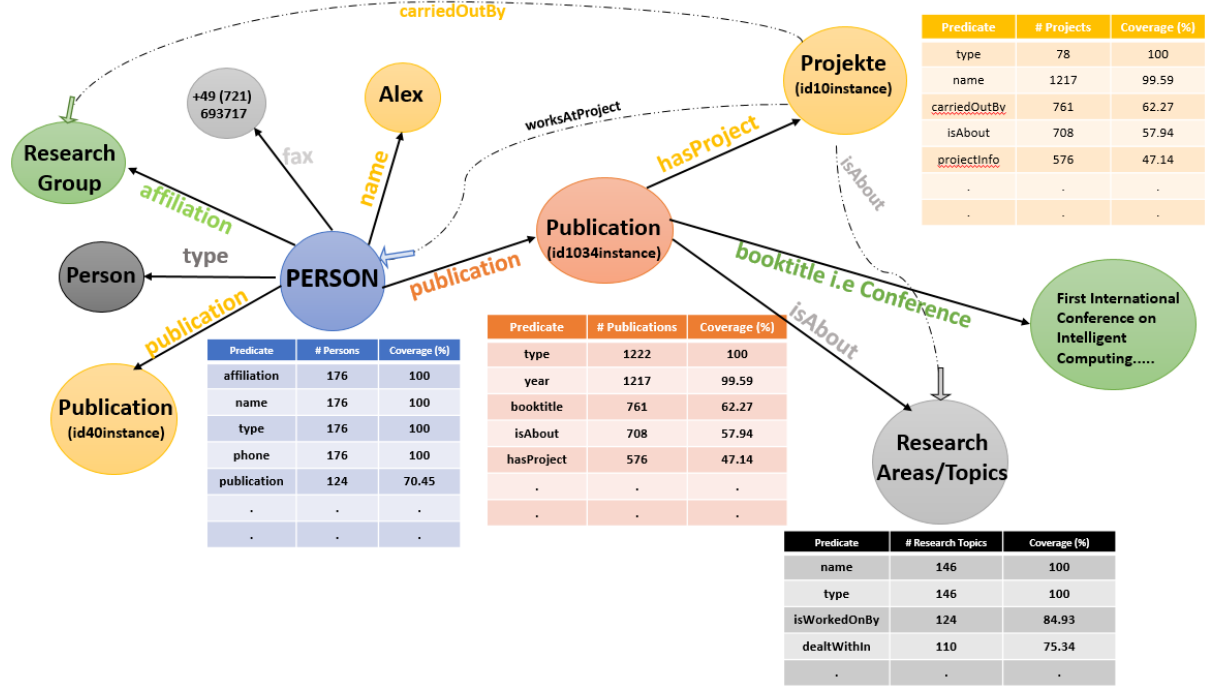


Figure 2: Simplified representation of the neighbourhood explored around each Person node. The figure highlights the main RDF entities, their relationships, and the coverage of the most informative predicates used during feature engineering.

The feature engineering process started by analysing the neighbourhood of the labelled Person nodes. Instead of converting every predicate directly into a machine learning feature, the surrounding entities were first explored to identify information that could help distinguish between research groups. Figure 2 summarises the main entities and relationships considered during this analysis.

The analysis showed that publications contain rich semantic information through research topics, conference venues and publication years. Projects provide additional information about research activities, while direct person attributes such as phone number and fax were found to have little predictive value. The coverage of the important predicates was also analysed to ensure that selected features were sufficiently represented across the dataset.

3.3 Feature Engineering Decisions

Table 2: Feature engineering strategy used to convert RDF data into a machine learning dataset.

Feature	Encoding	Reason
Homepage	Binary	Indicates presence of a homepage
Publication Count	Integer	Preserves publication activity
WorksAtProject	Count	Measures project involvement
Research Topics	Multi-hot	A researcher may work on multiple topics
Conference	Multi-hot	Captures publication venue
Publication Year	Integer	Represents temporal information
Number of Projects	Integer	Avoids project identifier leakage

During the analysis, project identifiers were carefully examined before being included as machine learning features. It was observed that approximately 95% of projects were associated with only one research group. Using project identifiers directly would therefore allow the model to memorize the correct class instead of learning meaningful patterns. To avoid this target leakage, only the number of projects associated with each researcher was retained as a numerical feature.

3.4 Final Feature Matrix

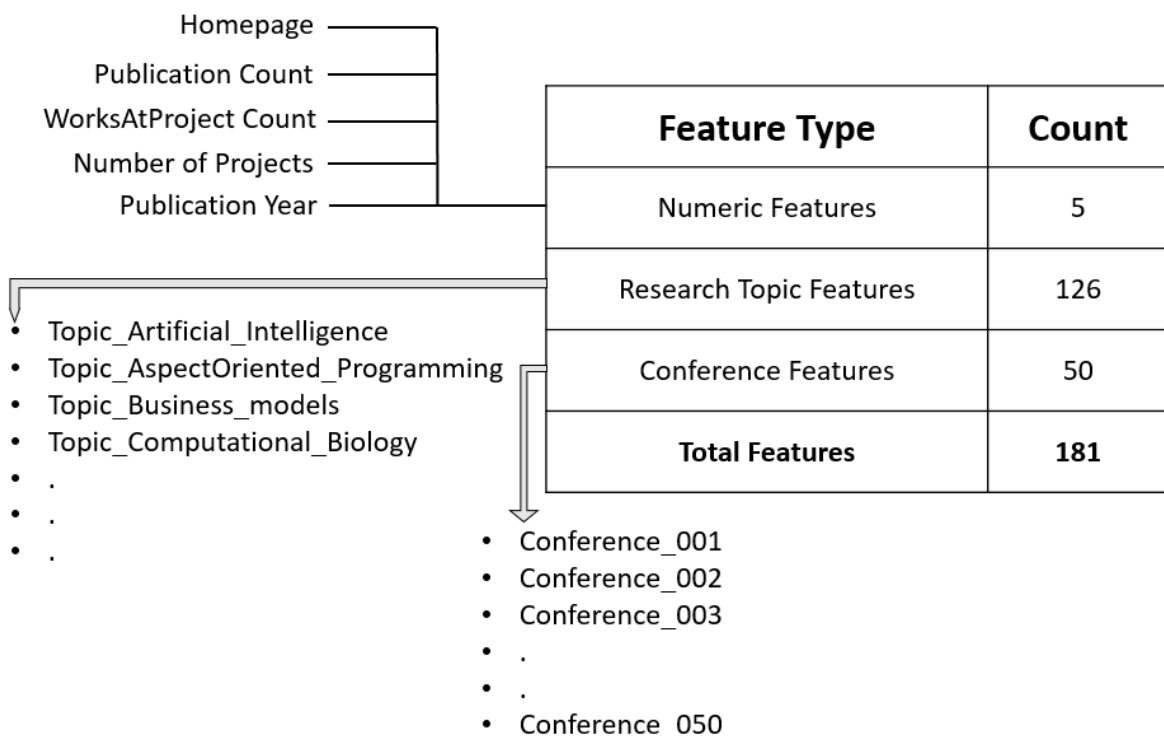


Figure 3: Overview of the final feature matrix used for model training.

After feature engineering, each researcher was represented by 181 engineered features consisting of numerical attributes, research topics and conference information. The final feature matrix contains 176 researchers with no missing values or duplicate entries and was used as the input for model training.

4 Model Training and Evaluation

4.1 Model Selection

After constructing the feature matrix, several supervised machine learning algorithms were trained to predict the research group of each researcher. The original training and testing split provided with the AIFB dataset was preserved throughout the experiments to ensure a fair comparison between models.

Four classical machine learning algorithms were selected for evaluation: Logistic Regression, Decision Tree, Random Forest, and XGBoost. These models were chosen because they represent different learning approaches, ranging from simple linear classifiers to ensemble-based tree methods. Model performance was evaluated using training accuracy, testing accuracy, macro precision, macro recall and macro F1-score.

4.2 Performance Comparison

Table 3: Performance comparison of the evaluated machine learning models.

Model	Train Acc.	Test Acc.	Macro Precision	Macro Recall	Macro F1
Logistic Regression	86.43	75.00	69.58	62.92	64.76
Random Forest	92.86	75.00	72.59	60.83	62.25
Decision Tree	92.86	72.22	61.62	59.17	59.77
XGBoost	92.86	72.22	55.56	55.00	54.92

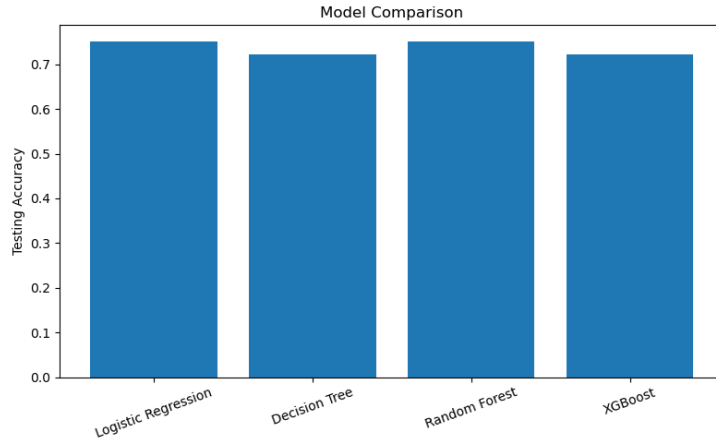


Figure 4: Comparison of the performance metrics for the evaluated machine learning models.

Table 3 compares the performance of the four evaluated machine learning models. Logistic Regression and Random Forest achieved the highest testing accuracy of 75%, while Decision Tree and XGBoost achieved 72.22%. Although Random Forest showed a higher training accuracy, the testing accuracy remained comparable to Logistic Regression, indicating a moderate level of overfitting. Overall, the differences in predictive performance between the evaluated models were relatively small.

4.3 Selected Model

Although Logistic Regression and Random Forest achieved the same testing accuracy, Random Forest was selected for the explainability analysis. Tree-based ensemble models are capable of

capturing non-linear relationships between features while remaining compatible with several explainability techniques discussed in the course, including Feature Importance, Permutation Feature Importance, Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE) plots, and SHAP. Therefore, Random Forest provides a suitable balance between predictive performance and interpretability for this project.

5 Model Explanation

5.1 Global Feature Importance

The first explanation technique used in this project is the built-in feature importance provided by the Random Forest classifier. This measure estimates how much each feature contributes to reducing impurity during tree construction. Features with higher importance values have a greater influence on the model's overall predictions.

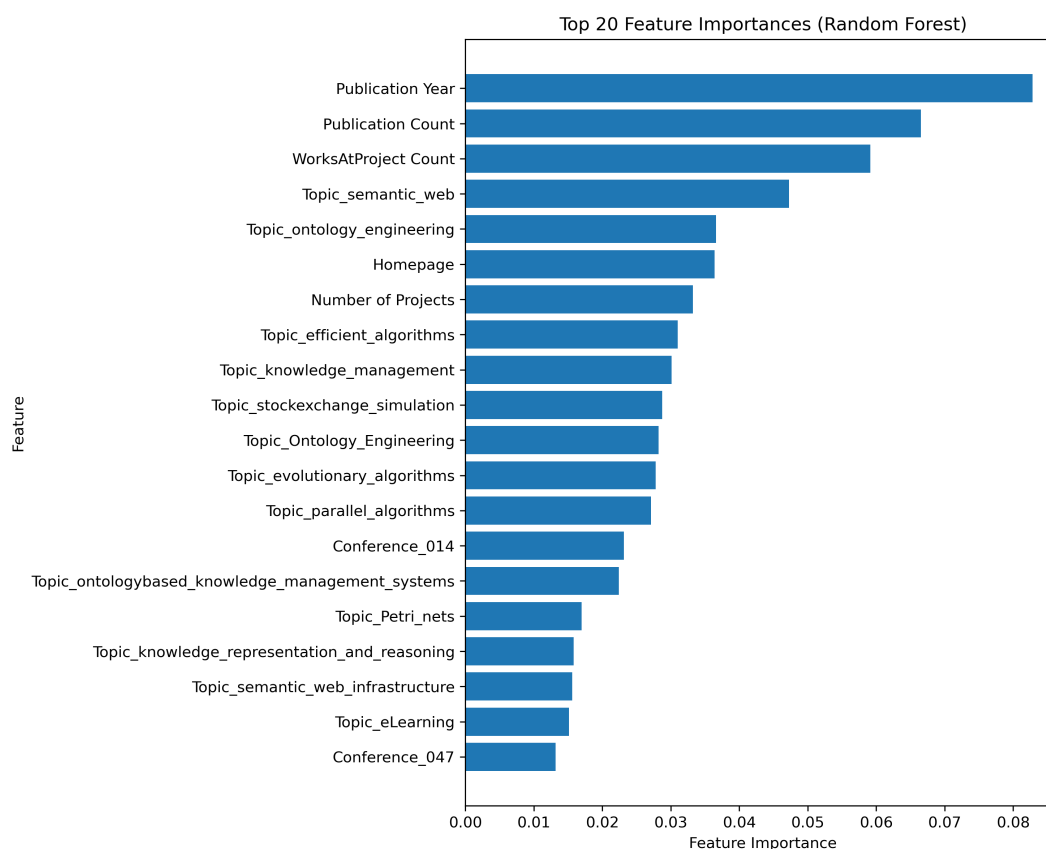


Figure 5: Top 20 feature importances obtained from the Random Forest classifier.

Figure 5 shows that Publication Year, Publication Count and WorksAtProject Count are the three most influential numerical features. Several research topics, such as semantic web, ontology engineering and knowledge management, also appear among the most important features. This indicates that both publication characteristics and research interests contribute to distinguishing research groups within the AIFB knowledge graph.

5.2 Permutation Feature Importance

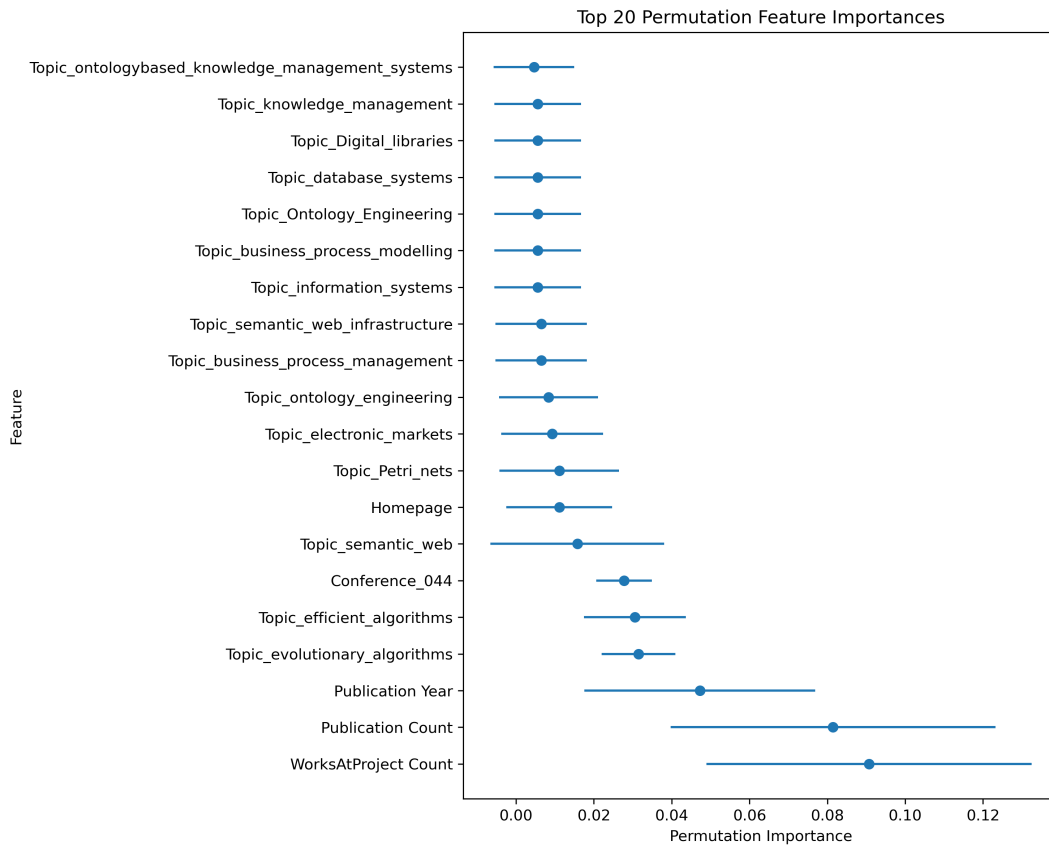


Figure 6: Top 20 features ranked using permutation feature importance.

Unlike built-in feature importance, permutation importance measures the decrease in prediction accuracy after randomly shuffling each feature. Features causing a larger decrease are considered more important because the model relies heavily on them.

As shown in Figure 6, WorksAtProject Count and Publication Count become the most influential features, followed by Publication Year. Several research topics also remain among the top-ranked features. Although the exact ranking differs slightly from the built-in importance, both methods consistently identify publication-related information and research topics as the main factors used by the model.

Table 4: Comparison of the ten most influential features according to Random Forest feature importance and permutation feature importance.

Feature	Random Forest	Permutation
WorksAtProject Count	0.059129	0.090741
Publication Count	0.066539	0.081481
Publication Year	0.082810	0.047222
Topic_evolutionary_algorithms	0.027772	0.031481
Topic_efficient_algorithms	0.031027	0.030556
Conference_044	0.011526	0.027778
Topic_semantic_web	0.047280	0.015741
Topic_Petri_nets	0.016970	0.011111
Homepage	0.036370	0.011111
Topic_electronic_markets	0.007975	0.009259

Both importance measures produce similar rankings, increasing confidence that the identified features genuinely influence the model instead of being artifacts of a particular importance metric.

5.3 Partial Dependence Plots

Partial Dependence Plots (PDP) illustrate the average effect of an individual feature on the model prediction while averaging over all remaining features. They provide a global interpretation of how changing one feature influences the predicted probability of a selected class.

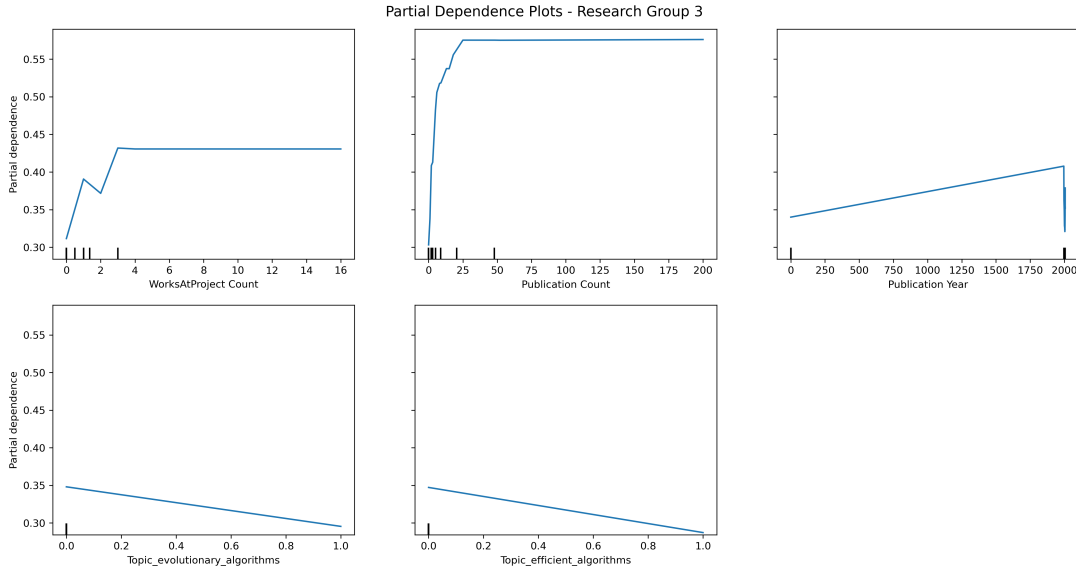


Figure 7: Partial dependence plot illustrating the influence of the selected feature on the predicted probability of Research Group 3.

Write one paragraph interpreting the trend observed in the partial dependence plot.

Figure 7 presents the PDPs for the five most important features with respect to Research Group 3. Increasing Publication Count generally increases the predicted probability until the effect stabilizes, suggesting that additional publications beyond a certain point provide limited additional information. WorksAtProject Count also shows a positive relationship before reaching a plateau. In contrast, the presence of the research topics "evolutionary algorithms" and "efficient

algorithms” slightly decreases the probability of belonging to Research Group 3, indicating that these topics are more characteristic of other research groups.

5.4 ICE Plots

While PDP shows the average behaviour across all researchers, Individual Conditional Expectation (ICE) plots visualize the prediction trajectory for each individual researcher separately. This allows the identification of heterogeneous responses that may be hidden by the average trend shown in PDP.

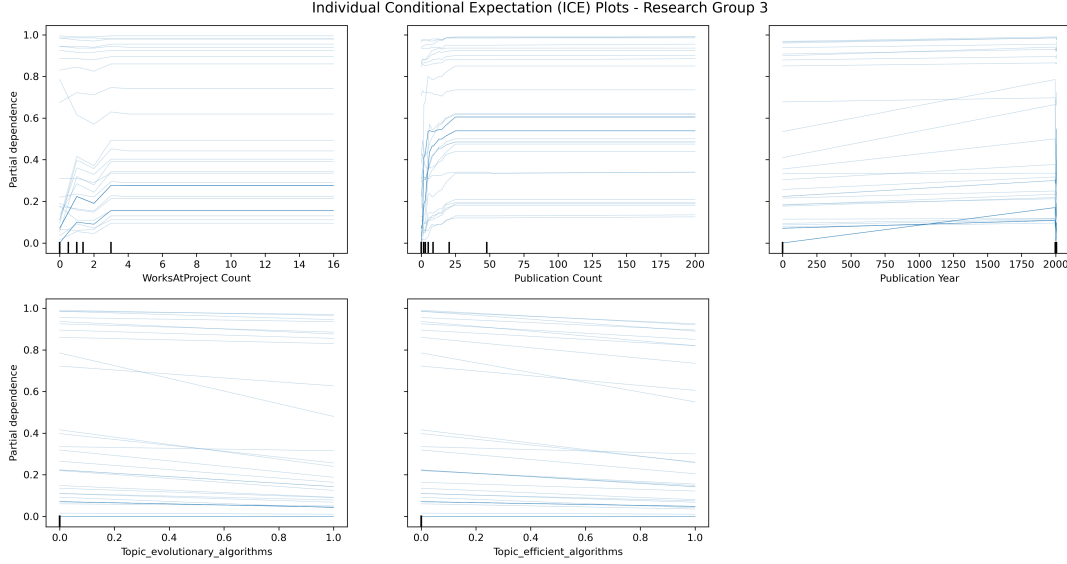


Figure 8: Individual Conditional Expectation (ICE) plot showing the effect of the selected feature on predictions for Research Group 3.

Figure 8 shows that most researchers follow trends similar to the PDP, although several individual prediction curves deviate noticeably. This indicates that the effect of some features depends on the characteristics of individual researchers rather than remaining identical across the dataset. Therefore, ICE complements PDP by revealing local variations in model behaviour.

5.5 SHAP Explanations

SHAP (SHapley Additive exPlanations) assigns every feature a contribution value for each prediction based on cooperative game theory. Unlike traditional feature importance methods, SHAP provides both global and local explanations while preserving consistency across features.

5.5.1 Global SHAP

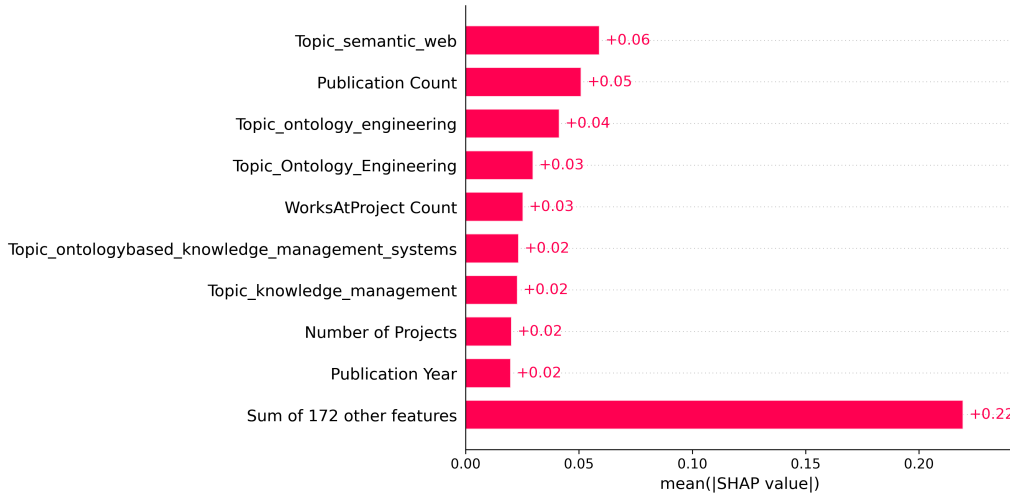


Figure 9: Global SHAP feature importance based on the mean absolute SHAP values.

Figure 9 shows the mean absolute SHAP values across the entire test dataset. `Topic_semantic_web` is identified as the most influential feature, followed by `Publication Count` and `Topic_ontology_engineering`. The ranking confirms the observations obtained from Feature Importance and Permutation Importance, indicating that publication characteristics together with research topics are the dominant factors influencing model predictions.

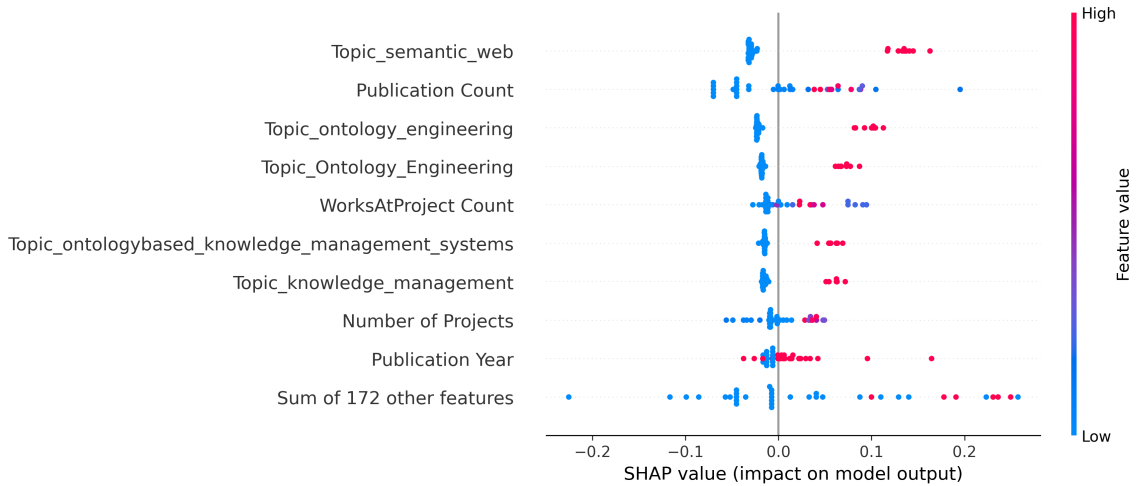


Figure 10: SHAP beeswarm plot illustrating the distribution and impact of feature contributions across all samples.

The SHAP beeswarm plot in Figure 10 provides additional insight into how feature values influence predictions. Each point represents one researcher, while the horizontal position indicates the feature's contribution to the model output. Positive SHAP values increase the probability of a class prediction, whereas negative values decrease it. The distribution shows that high values of `Topic_semantic_web` and `Publication Count` frequently contribute positively to model predictions, while lower values often have the opposite effect.

5.5.2 Local SHAP

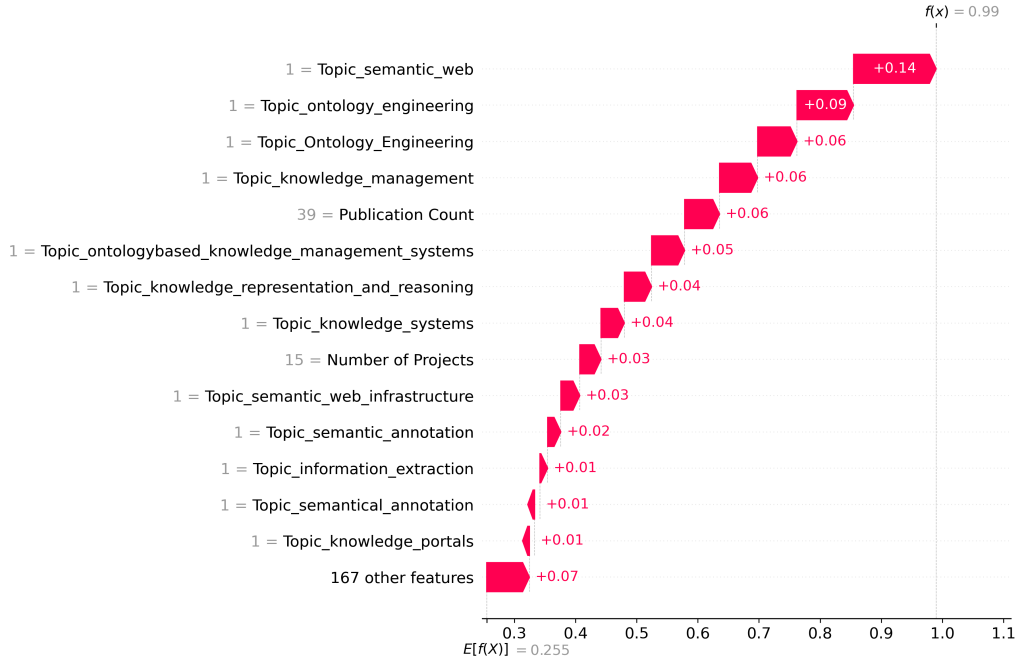


Figure 11: SHAP waterfall plot explaining a correctly classified instance.

Figure 11 presents the SHAP explanation for a correctly classified researcher belonging to Research Group 3. The prediction confidence is approximately 99%, indicating a highly confident classification. The largest positive contributions originate from Topic_semantic_web, Topic_ontology_engineering, Publication Count and several knowledge-management related topics. Together, these features increase the prediction from the expected model output to the final probability of the predicted research group.

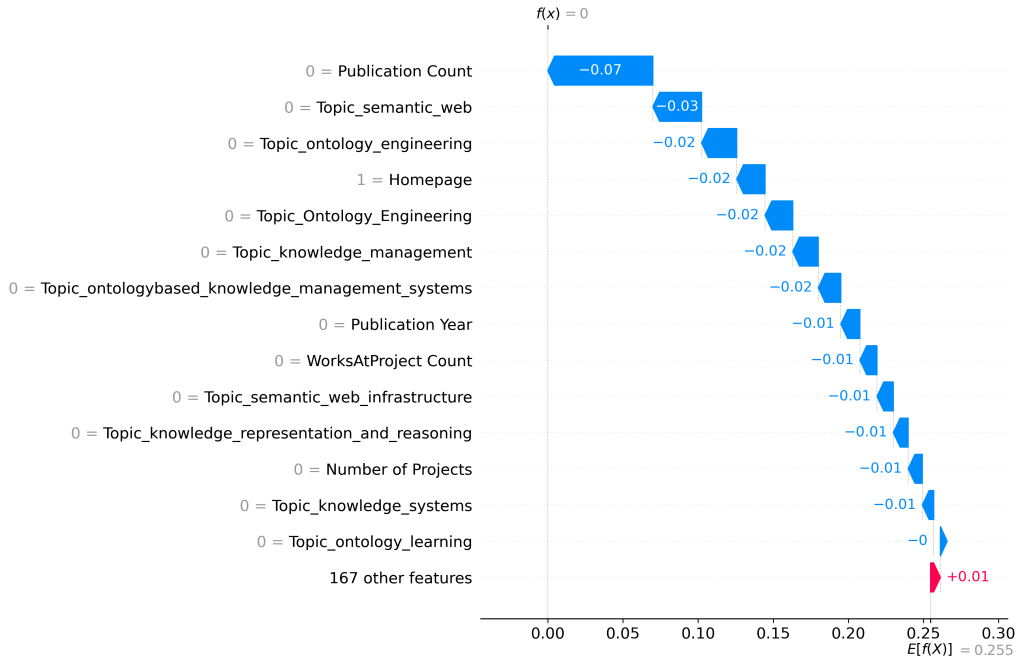


Figure 12: SHAP waterfall plot explaining an incorrectly classified instance.

Figure 12 illustrates an incorrectly classified researcher. Although the actual label belongs to Research Group 2, the model predicts Research Group 1 with approximately 78% confidence. The explanation shows that Publication Count, semantic web related topics and ontology engineering dominate the decision, suggesting that the researcher shares characteristics commonly observed within Research Group 1. This example demonstrates how SHAP helps understand model errors by identifying which features led to an incorrect prediction rather than treating the prediction as a black box.

6 Conclusion

In this project, we developed an explainable machine learning pipeline for predicting the research group affiliation of researchers using the AIFB RDF knowledge graph. Starting from the raw RDF graph, we explored the relationships between different entities and carefully designed a tabular feature representation by selecting informative features while avoiding target leakage. Several classification models were evaluated, and Logistic Regression and Random Forest achieved the best overall performance with a testing accuracy of 75%. To understand the model behaviour, multiple explainability techniques including Feature Importance, Permutation Feature Importance, Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE) plots and SHAP were applied. These explanations showed that publication-related features together with research topics were the most influential factors in the predictions and also helped explain both correct and incorrect classifications. Overall, this project demonstrates that combining careful feature engineering with explainable AI techniques makes machine learning models trained on RDF knowledge graphs more transparent and easier to interpret.

7 Contributions of Team Members

Table 5: Contributions of the project team members.

Team Member	Contribution
Manoj Billur Nagen-dra Prasad	RDF exploration, feature engineering, Random Forest and XGBoost model implementation and evaluation, Partial Dependence Plot (PDP) and Individual Conditional Expectation (ICE) analysis, SHAP-based global and local explanations, notebook organization, GitHub repository management, and report preparation.
Steffi Stephen	Dataset selection, discussion of the feature engineering strategy, implementation and evaluation of Logistic Regression and Decision Tree models, Global Feature Importance and Permutation Feature Importance analysis, and review of intermediate results.

The overall project design, including dataset selection, feature engineering strategy and interpretation of the results, was discussed jointly throughout the project. Individual implementation tasks were divided between the team members as listed above, while the final report was prepared by combining the contributions from both members.

Acknowledgement

We would like to acknowledge the resources that supported the completion of this project. The lecture material, tutorial exercises, and classroom discussions provided the primary foundation for understanding the explainable AI techniques used throughout this work.

For the data analysis and feature engineering stages, we referred to publicly available online resources and examples to better understand how RDF knowledge graphs can be explored and transformed into machine learning datasets. However, the final feature selection, encoding decisions, leakage analysis, and overall feature engineering strategy were developed based on our own analysis of the AIFB dataset.

During model implementation and evaluation, we followed the approaches introduced in the lecture and tutorial exercises and referred to the official scikit-learn and XGBoost documentation for the correct usage of the respective machine learning algorithms and evaluation methods.

For the explainability part of the project, we implemented the techniques presented in the course, including Feature Importance, Permutation Feature Importance, Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE) plots, and SHAP. ChatGPT was additionally used as a learning assistant to clarify concepts, discuss implementation ideas, improve code organization, and assist in structuring and proofreading the report. All implementation decisions, experimental results, interpretations, and conclusions presented in this report were reviewed and finalized by the authors.